

Muscle Fatigue Detection Using Sparse Ultrasound and EMG

ES 227 Final Report

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Abstract

Muscle fatigue is associated with reduced force-generating capacity, altered movement quality, and increased risk of muscle injury (Dugan & Frontera, 2000; Enoka & Duchateau, 2008), yet it remains difficult to quantify objectively during real-world training or rehabilitation. Current approaches, such as surface electromyography, force measurement, near-infrared spectroscopy, and conventional ultrasound each capture only part of the fatigue process and are often limited by noise, laboratory constraints, or lack of muscle-specific mechanical information (Li et al., 2024). This project developed and evaluated an algorithmic fatigue assessment pipeline intended to support a future wearable system that combines sparse A-mode ultrasound with EMG and torque validation. The system focuses on the biceps brachii and uses two main feature pathways: ultrasound-derived echogenicity changes within a muscle region of interest and electromechanical delay between EMG activation and ultrasound-detected mechanical response (Alvarez et al., 2024; Smith et al., 2017). A standardized protocol collected synchronized ultrasound, EMG, and dynamometer data during repeated maximum voluntary isometric contractions and fatiguing exercise trials. This data showed that echogenicity increased as MVIC decreased across trials, with an adjusted R^2 of 0.9197 for the reported polynomial fit. EMD analysis showed that two of three subjects followed the expected trend in which ultrasound-derived delay changed consistently with the EMG-to-torque baseline, while one subject showed an inconsistent trend likely related to boundary tracking error. These results support the feasibility of ultrasound and EMG-based fatigue monitoring, while also highlighting the need for additional participants, improved boundary tracking, controlled rest intervals, and wearable integration before the approach can be generalized.

1. Introduction

1.1 Problem Definition and Motivation

Muscle fatigue is commonly understood as a decline in a muscle's ability to generate force or power during sustained or repeated activity (Enoka & Duchateau, 2008). In sports medicine and rehabilitation, this decline matters because fatigued muscle can reduce performance, alter movement quality, and increase susceptibility to overload or strain (Dugan & Frontera, 2000; Schwiete et al., 2025). The project materials frame the central problem as follows: fatigue is clinically important, but it is difficult to measure in a way that is objective, muscle-specific, and practical outside a laboratory (Li et al., 2024). The team therefore focused on an algorithmic approach that could eventually be deployed in a wearable form factor for athletic training, rehabilitation, and ergonomics.

The design opportunity is especially clear in settings where users must decide how hard to train or rehabilitate without overloading tissue. Exercise scientists, clinicians, and trainers often rely on indirect signs such as perceived exertion, technique breakdown, repetition failure, or loss of force output. These measures can be useful, but they do not directly reveal local muscle mechanical state. The proposed system targets peripheral fatigue by measuring local biceps brachii changes with sparse ultrasound while using EMG and torque as validation signals (Alvarez et al., 2024; Jin et al., 2024).

1.2 Physiological Background

Fatigue involves both central and peripheral mechanisms (Taylor et al., 2016). Central fatigue reflects reductions in neural drive or altered motor unit recruitment (Amann & Dempsey, 2008), whereas peripheral fatigue includes impaired excitation-contraction coupling, metabolite accumulation, calcium handling changes, cross-bridge dysfunction (Nocella et al., 2011), and reduced musculotendinous stiffness. During sustained or repeated maximal voluntary contractions, these mechanisms can produce measurable loss of force output and delayed mechanical response (Smith et al., 2017). This project uses those physiological effects to motivate two sensing strategies: echogenicity analysis, which targets changes in ultrasound backscatter from muscle tissue (Shi et al., 2007), and

electromechanical delay analysis, which targets the time lag between electrical activation and mechanical deformation (Smith et al., 2017; Alvarez et al., 2024).

Fresh muscle produces a more immediate and consistent mechanical response, while a fatigued muscle shows altered tissue response and delayed contraction dynamics (Smith et al., 2017). For the paper, this section should connect the physiology directly to the algorithm: if fatigue changes local tissue structure and slows force transmission, then ultrasound and EMG can provide complementary measurements of the same underlying process (Alvarez et al., 2024; Shi et al., 2007).

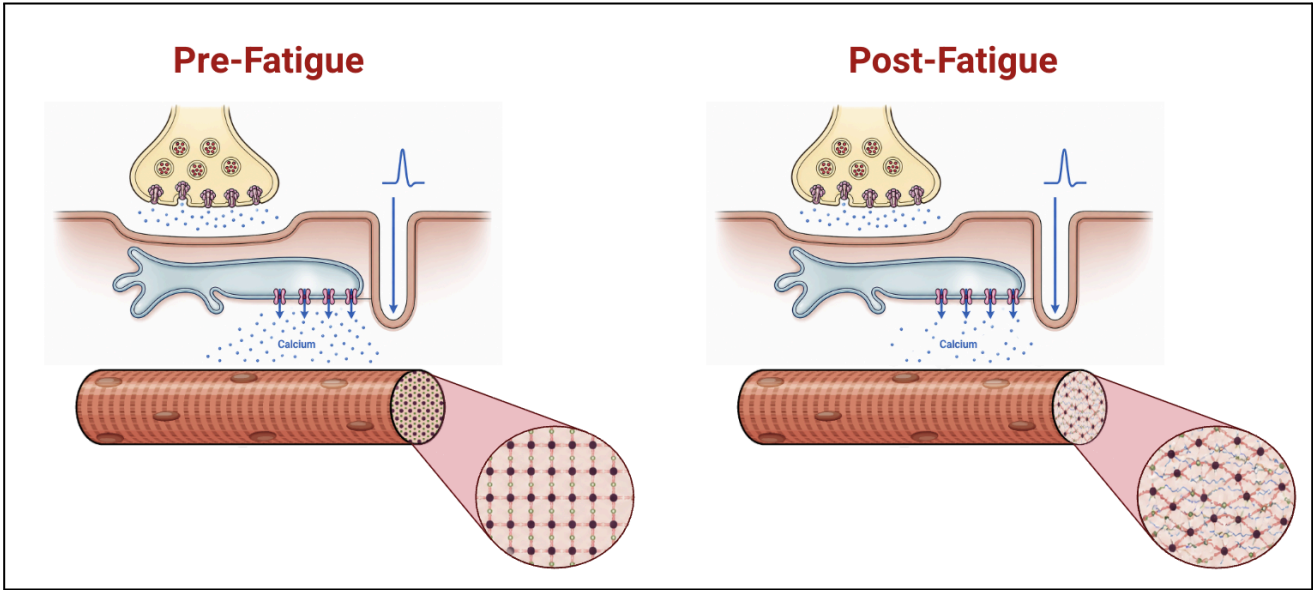


Figure 1. Simplified physiology diagram showing pre-fatigue vs. post-fatigue muscle response

1.3 Stakeholders and Use Environment

Stakeholder	Role	Needs / Priorities
Exercise scientists and strength coaches	Use fatigue information to guide load, intensity, and recovery decisions.	Actionable zones, rapid feedback, repeatable measurements, low disruption to training.
Athletes	Perform repeated training sets while avoiding overload-related injury.	Comfort, minimal movement interference, clear feedback, safety.
Physical therapists / rehabilitation clinicians	Progress strengthening while avoiding reinjury or excessive fatigue.	Objective metrics, patient-specific baselines, clinical interpretability, workflow fit.
Researchers and device developers	Validate fatigue mechanisms and develop wearable systems.	Raw synchronized data, reproducible algorithms, validation against force and EMG.

1.4 Mission Statement

Problem statement: Exercise scientists have no practical method to assess muscle fatigue while guiding training intensity, leading to suboptimal performance and injury.

Final mission statement: Exercise scientists need algorithms to assess muscle fatigue in a wearable form factor, leveraging EMG and sparse ultrasound that guide training intensity.

2. Background Research and Prior Art

2.1 Existing Fatigue Measurement Approaches

Several non-invasive fatigue-monitoring approaches have been studied but all fall short of what is needed. Surface EMG is widely used to measure muscle activation and fatigue-related spectral shifts (Stulen & De Luca, 1982), but it is vulnerable to sweat, skin impedance changes, electrode placement variability, motion artifact, crosstalk, and difficulty sensing deeper muscles (Li et al., 2024). Near-infrared spectroscopy (NIRS) provides metabolic information through muscle oxygenation but is limited by tissue depth, latency, and difficulty establishing universal thresholds (Tuesta et al., 2022). Force or torque measurement provides a direct mechanical benchmark, but it usually requires external instrumentation and does not by itself explain local muscle physiology. Conventional B-mode ultrasound can visualize architecture, but it is often bulky and computationally expensive for continuous wear (Qu et al., 2024).

A-mode ultrasound was selected as a promising middle ground because it provides a one-dimensional depth signal that can track local muscle mechanical behavior with lower hardware and computational burden than full imaging ultrasound (Jin et al., 2024; Xue et al., 2023). Prior work using sparse or wearable ultrasound motivated the team's focus on biceps brachii boundary tracking, echogenicity, and deformation dynamics (Alvarez et al., 2024; Qu et al., 2024). The project therefore does not claim to have built a fully finished wearable device; instead, it develops and validates the algorithmic foundation that a wearable A-mode ultrasound and EMG device could use.

Approach	What it measures	Strengths	Limitations motivating this project
sEMG	Electrical activity and motor unit behavior	Non-invasive, common in fatigue studies, compatible with wearable electrodes	Sensitive to placement, sweat, crosstalk, motion artifact, and signal degradation.
NIRS	Muscle oxygenation and metabolic demand	Real-time non-invasive metabolic context	Limited depth, relative rather than absolute measures in common systems, does not directly measure mechanics.
Force / torque	Mechanical output	Clear ground truth for performance decline	Requires dynamometer or load cell and cannot isolate internal muscle mechanisms alone.
B-mode ultrasound	2D muscle architecture	Rich structural information	Bulky and computationally demanding for continuous use.
Sparse A-mode ultrasound	Depth-dependent tissue signal and deformation	Lower data burden, wearable potential, muscle-specific mechanical signal	Requires robust placement, ROI selection, boundary tracking, and validation.

2.2 Patents and Commercial Systems

Existing fatigue-monitoring solutions generally fall into two categories, each with a key limitation. The first category measures analogs of fatigue — physiological signals such as heart rate, skin temperature, or oxygen saturation that correlate with overall exertion but do not reflect the mechanical state of a specific muscle (Li et al., 2024; Hwang et al., 2025). Wearable systems based on remote physiological sensing and high-density electrode garments fall into this group; they capture global effort or aggregate muscle activity, but cannot resolve whether a particular muscle is fatigued.

The second category measures fatigue more directly but relies on instrumentation that is not wearable. B-mode ultrasound captures muscle architecture and echogenicity with high fidelity, but requires bulky probes and benchtop processing (Shi et al., 2007). Electromechanical delay measurements have been validated against dynamometer-derived torque (Smith et al., 2017), but the dynamometer itself constrains the user to a fixed laboratory setup. Commercial systems such as spectral-EMG fatigue monitors (Stulen & De Luca, 1982) and quadriceps force-output devices (Xue et al., 2023) similarly trade portability for signal quality, or vice versa. The gap, then, is a system that measures muscle-specific fatigue directly, using mechanical and neural signals from the muscle itself, in a form factor practical enough for real-world training and rehabilitation. This project targets that gap by combining sparse A-mode ultrasound with EMG, two modalities whose hardware footprint is compatible with a future wearable (Alvarez et al., 2024; Jin et al., 2024; Qu et al., 2024).

3. Design Specifications and Concept Selection

3.1 Functional Requirements

User Need	Functional Requirement	Quantitative Target
Detect peripheral muscle fatigue	Multimodal fatigue prediction model shall estimate reduction in Maximum Voluntary Contraction (MVC)	Cross-validated $R^2 \geq 0.85$ and $MAE \leq 10\%$ MVC across participants
	Fatigue metric shall be within acceptable deviation of ground truth fatigue	$\pm 10\%$ of MVC reduction (or defined % threshold)
	Supervised classification model shall categorize fatigue into discrete physiological zones	≥ 5 fatigue levels with $\geq 80\%$ accuracy and macro-F1 ≥ 0.80 vs MVC-based stratification
Function in real-world clinical or sports settings	Device shall allow correct placement by minimally trained user	$\geq 95\%$ correct placement after ≤ 5 min training; feedback if incorrect
	Signal shall be robust to mechanical perturbation	$\leq 10\%$ signal drift under ≤ 50 N compressive force

	System shall process fatigue metric within hardware constraints, on board	Runs on onboard processor with $\leq 80\%$ CPU utilization and $\leq X$ MB memory
Maintain stable fatigue estimation during prolonged wear	Model performance shall remain consistent despite gradual signal-quality changes over time	$\leq 10\%$ degradation in R^2 or classification accuracy over 2-hour wear session

3.2 Concepts Considered

The team considered three main algorithmic strategies. The first was a multimodal machine-learning model combining A-mode ultrasound, EMG, and IMU features to estimate fatigue as a personalized Muscle Fatigue Index (Hwang et al., 2025; Xue et al., 2023). The second was an echogenicity-based method using ultrasound backscatter changes within a region of interest as a mechanical fatigue proxy (Shi et al., 2007; Qu et al., 2024). The third was an electromechanical delay method using the time between EMG activation and ultrasound-detected mechanical response as a fatigue-sensitive timing feature (Smith et al., 2017; Alvarez et al., 2024). Over the semester, the final design evolved from a single-sensor wearable concept into an algorithmic pipeline that evaluates echogenicity and EMD as separate but complementary fatigue features, with torque serving as the mechanical validation signal.

Concept	Core idea	Advantages	Main concerns
Multimodal ML fatigue index	Fuse A-mode ultrasound, EMG, and IMU features to estimate MVC/MVIC decline or fatigue zones.	Most expandable; can learn nonlinear relationships and down-weight noisy modalities.	Requires larger dataset, risk of overfitting, more compute, harder interpretation.
A-mode/M-mode echogenicity	Measure ROI ultrasound signal intensity or relative echogenicity changes during fatigue.	Computationally efficient, directly related to muscle tissue response, compatible with wearable ultrasound.	Sensitive to ROI selection, probe pressure, boundary tracking, and interpretation direction.
Electromechanical delay	Measure delay between EMG onset and ultrasound mechanical onset, validated against EMG-to-torque delay.	Physiologically interpretable timing metric; uses complementary neural and mechanical signals.	Requires synchronization, robust onset detection, and accurate muscle boundary tracking.

3.3 Concept Selection

To compare the three candidate fatigue-detection strategies, the team used a 1-3 scoring matrix, where 3 indicates the strongest performance for a given criterion and 1 indicates the weakest performance. The criteria were selected to evaluate both technical feasibility and eventual usability, including computational burden, interpretability, comfort, real-time potential, ability to quantify fatigue, setup simplicity, correlation with MVIC, robustness, and placement sensitivity. The matrix was used as a relative design tool rather than a final validation result.

Criterion	ML Correlation	Echogenicity	EMD
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Fatigue-index quality and MVC correlation	3	2	2
Clinical interpretability of fatigue output	3	1	2
Real-time implementation feasibility	2	1	3
Sensor wearability and setup burden	1	3	2
Robustness to perturbation and signal variability	3	1	2
Total	12	8	11

The initial scoring favored the multimodal ML strategy, with EMD as the next strongest concept and echogenicity as the simplest ultrasound-only pathway. In practice, the final project adopted a staged strategy: build interpretable echogenicity and EMD pipelines first, validate them against MVIC/torque behavior, and use those features as the basis for a future fatigue-zone classifier or wearable fatigue index.

ML Correlation scored highest overall because it has the greatest potential to combine multiple fatigue-related features into a single quantitative fatigue index. Although machine-learning models are often less transparent than direct measurements, the intended ML approach would use defined features from ultrasound and EMG rather than raw black-box inputs. Therefore, “signal interpretability” refers to the interpretability of the final fatigue output relative to MVIC decline, not the inherent simplicity of the algorithm. Echogenicity was scored lower because ultrasound brightness can be affected by probe pressure, gain, region-of-interest selection, and motion, making it difficult to interpret as fatigue on its own. EMD scored between the two because it has a clear physiological meaning, the delay between electrical activation and mechanical contraction, but still depends on accurate onset detection in both EMG and ultrasound signals.

Because several scores depend on the assumed future hardware configuration, the matrix should be interpreted as a guide for design direction rather than proof that one method is definitively superior.

4. Final System Design and Implementation

4.1 System Overview

The final system is best described as an algorithmic fatigue assessment platform rather than a completed consumer wearable. The long-term vision is a wearable device using sparse ultrasound and EMG to report fatigue zones after or during training sets. The semester prototype used laboratory instrumentation to collect synchronized data, process ultrasound and EMG signals, and test whether the resulting features followed physiologically meaningful fatigue trends.

Subsystem	Role in Prototype	Role in Future Wearable
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Sparse A-mode / M-mode ultrasound	Collects depth-dependent biceps signal and enables ROI echogenicity and boundary tracking.	Primary local mechanical sensing modality.
EMG system	Captures neuromuscular activation timing for EMD analysis.	Electrical activation reference and complementary fatigue feature.
Dynamometer / torque measurement	Provides mechanical output and MVIC benchmark.	Ground-truth validation during development; may not be part of final wearable.
Algorithm pipeline	Synchronizes signals, tracks muscle boundary, extracts features, detects onset, computes trends.	Transforms raw wearable sensor data into fatigue index or fatigue zones.

4.2 Instrumentation

The experimental setup combined an Optel ultrasound system, Delsys EMG equipment, a Delsys trigger module, an ADInstruments PowerLab system, and a dynamometer output. The PowerLab channels included a synchronization trigger, torque, position, and velocity. Ultrasound and EMG streams were aligned using the trigger signal so that muscle activation, tissue response, and torque output could be analyzed within the same MVIC window.

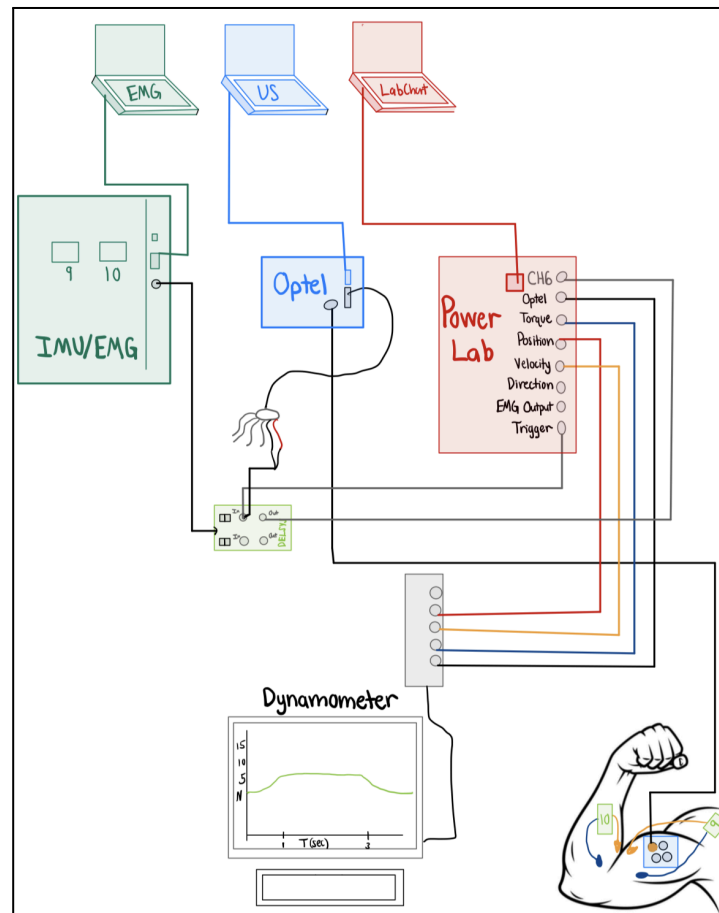


Figure 3. Instrumentation photo or wiring diagram.

4.3 Experimental Protocol

The experimental protocol was designed to synchronize torque, EMG, and A-mode ultrasound measurements during repeated biceps fatigue trials. Before each trial, the ultrasound transducer was positioned over the biceps brachii to capture a stable muscle region, and the EMG node was placed over the biceps to record electrical activation. The participant was instructed to keep the arm position consistent across trials and to follow the same sequence of rests, torque holds, curls, and maximum voluntary isometric contractions. The protocol began with calibration holds: a 5 Nm hold, 10s rest, 10 Nm hold, 10s rest, three MVIC attempts, 10s rest, another 5 Nm hold, 10s rest, and another 10 Nm hold. These known 5 Nm and 10 Nm holds were used to calibrate the voltage-to-torque conversion and check whether the dynamometer signal was stable. The three MVIC attempts were included so the strongest maximal contraction could be used as the subject's reference effort, consistent with standard MVIC assessment practices in fatigue research (Enoka & Duchateau, 2008).

After calibration, the participant completed four fatiguing sets. Each set followed the same structure: 10s rest, 5 Nm hold, 10s rest, 10 Nm hold, 10s rest, 10 concentric biceps curls with 2s holds, one MVIC, 10s rest, 5 Nm hold, 10s rest, and 10 Nm hold. The curl sequence was used to induce local biceps fatigue, while the MVIC provided the strongest standardized contraction for comparing fatigue-related changes across trials (Smith et al., 2017). The post-curl, post-MVIC, and end-rest windows were later used for echogenicity analysis (Shi et al., 2007; Qu et al., 2024), while the MVIC contraction was used for EMD onset detection because it produced the clearest EMG and mechanical response. Repeating this same structure across sets allowed the team to compare whether ultrasound-derived features and EMD changed as mechanical capacity declined.

Protocol Element	Purpose in Analysis
Opening rest	Baseline signal quality and drift check.
5 Nm / 10 Nm holds	Torque calibration and sanity check for voltage-to-torque conversion.
Fatiguing contractions/curls	Induce local fatigue and dynamic muscle response.
MVIC	Standardized high-effort contraction used for the strongest fatigue signal and onset analysis.
Post-MVIC / end rest windows	Windowed echogenicity comparisons and recovery/fatigue trend assessment.

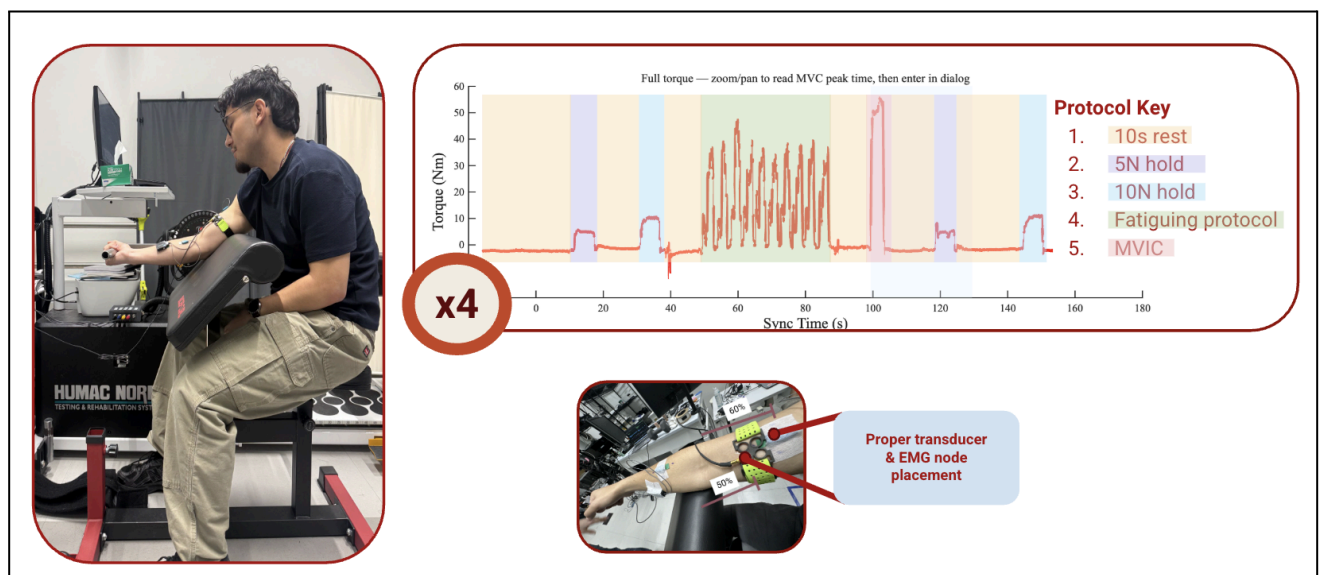


Figure 4. Experimental protocol schematic with colored windows labeled.

4.4 Ultrasound Boundary Tracking and Region of Interest

Because A-mode ultrasound measures a single depth line rather than a full image, selecting the correct muscle region is critical (Jin et al., 2024). The team used B-mode imaging to understand biceps anatomy and M-mode ultrasound to analyze repeated A-lines over time. The region of interest was bounded superficially at approximately 5 mm and extended to a deep biceps brachii boundary estimated by a moving boundary tracking algorithm. This boundary tracking step was important because muscle position and thickness can shift during contraction and fatigue (Shi et al., 2007; Alvarez et al., 2024); a fixed A-mode muscle region of interest (ROI) can include non-muscle tissue or miss the target tissue if the probe moves or the muscle deforms.

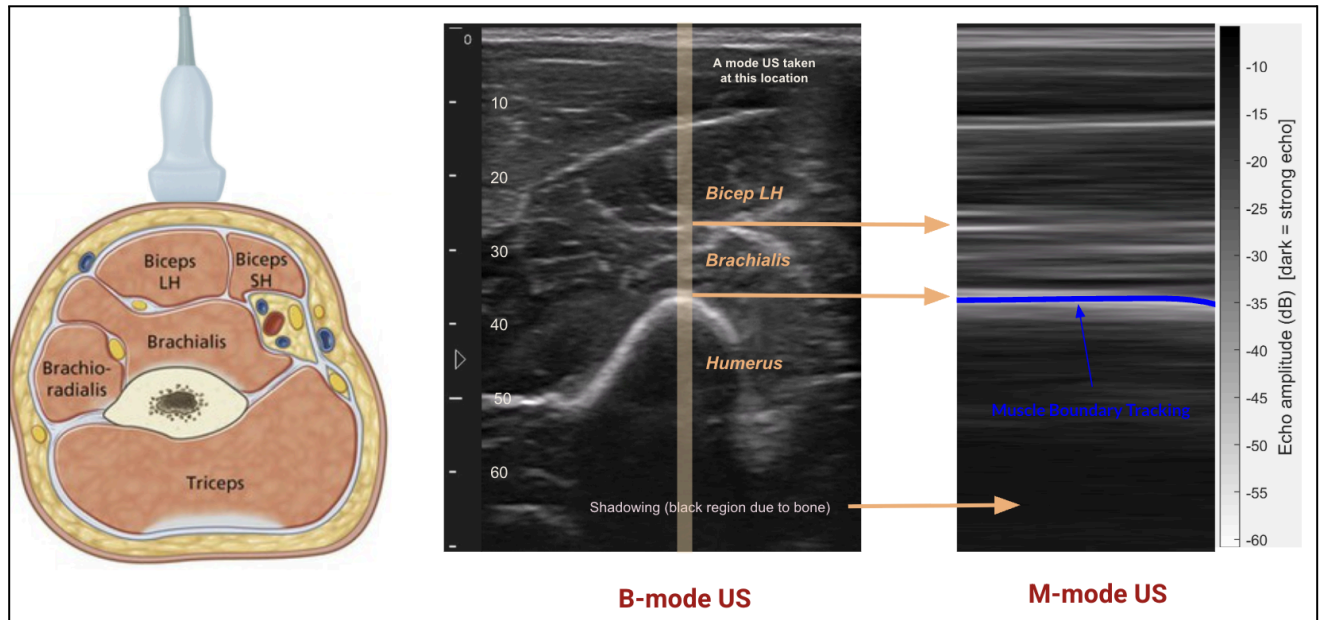


Figure 5. B-mode / M-mode comparison showing biceps brachii, humerus shadow, A-mode sampling location, and ROI boundary.

4.5 Echogenicity Pipeline

The echogenicity pipeline uses ultrasound signal intensity within the selected muscle ROI as a fatigue-sensitive mechanical feature, following approaches described in Shi et al. (2007) and Qu et al. (2024). The raw ultrasound signal is converted into an envelope representation, displayed as an M-mode image, and processed within the muscle boundaries. The pipeline selects the window, tracks the muscle boundary with MBTA, and computes ROI echogenicity. In the final analysis, mean ROI echogenicity was compared to MVIC across trials to determine whether ultrasound backscatter changed as force capacity declined (Alvarez et al., 2024).

Step	Description
Select analysis window	Choose a time interval around the relevant MVIC or rest window.
Process ultrasound envelope	Convert RF signal to envelope / M-mode representation for analysis.
Track muscle boundary	Use MBTA to identify the lower biceps boundary; use an assumed superficial boundary around 5 mm.
Compute ROI echogenicity	Average signal intensity within the ROI and compare across trials.

Relate to fatigue

Plot mean ROI echogenicity versus MVIC or trial number.

4.6 Electromechanical Delay Pipeline

The EMD pipeline estimates the delay between electrical activation and mechanical response, a feature previously validated as a marker of muscle fatigue (Smith et al., 2017; Alvarez et al., 2024). First, the three data streams are synchronized using a trigger. Second, each signal is processed with a modality-specific onset method. Torque onset is defined as the point where torque crosses 10% of the highest peak in the MVIC window. EMG onset is detected by finding the dominant burst and tracing backward to avoid false detection from earlier noise (Stulen & De Luca, 1982). Ultrasound onset is detected from the slope of the MBTA-derived mechanical response, rather than from a fixed threshold, because the ultrasound signal can sit at an elevated baseline before contraction (Alvarez et al., 2024). Finally, the pipeline compares EMG-to-torque delay and EMG-to-ultrasound delay across trials and subjects.

Signal	Processing / Onset Method	Reason
Torque	First crossing above 10% of MVIC peak within the target window.	Avoids small noise triggers while capturing mechanical force onset.
EMG	Bandpass and notch filtering, waveform-length envelope, backward-from-peak onset detection.	Anchors detection to the true activation burst and avoids false early noise spikes.
Ultrasound	Hilbert envelope / M-mode, MBTA boundary tracking, slope-based onset detection.	Detects mechanical deformation onset when baseline level is already elevated.

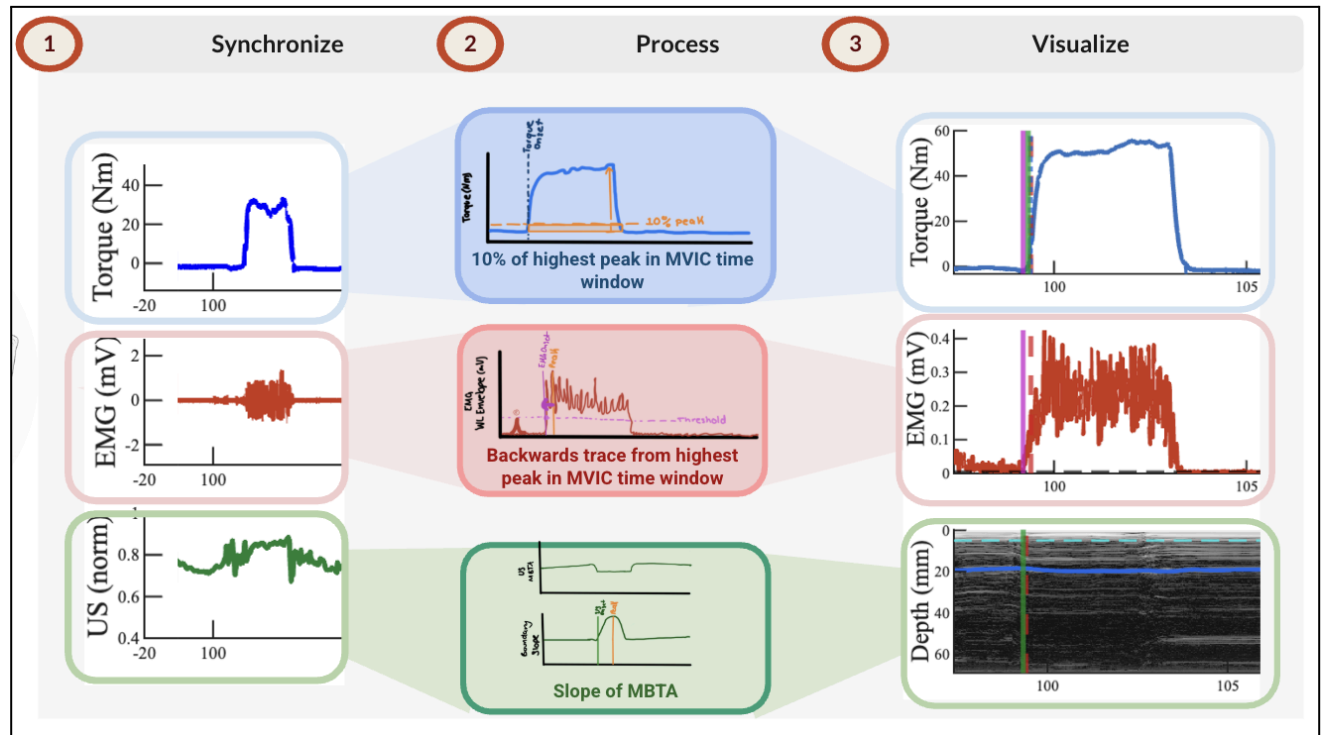


Figure 6. EMD pipeline showing synchronization, processing, and visualization.

5. Verification and Validation

5.1 Verification Plan

Requirement / Question	Verification Method	Current Evidence	Status
Can ultrasound features track fatigue-related force decline?	Compare mean ROI echogenicity or ERC to MVIC across trials.	Final deck shows mean ROI echogenicity increasing as MVIC decreases, Adj $R^2 = 0.9197$.	Promising, needs more subjects.
Can ultrasound-based EMD track mechanical delay?	Compare EMG-to-ultrasound delay trend to EMG-to-torque baseline.	Two of three subjects showed expected within-subject trend; one subject reversed.	Partially met.
Can data streams be synchronized?	Use trigger signal and inspect aligned torque, EMG, and ultrasound windows.	Weekly decks show synchronization and onset-detection workflows.	Working but still a limitation.
Can muscle ROI be tracked?	Use MBTA and B-mode-informed ROI selection.	Final deck uses boundary-tracked ROI; notes identify boundary tracking as a likely error source.	Functional but not robust enough.

5.2 Echogenicity Results

The final presentation reports an echogenicity trend across trials in which MVIC decreased while mean ROI echogenicity increased. In the displayed trials, MVIC declined from 45 to 32 across Trials A-D, while mean ROI echogenicity increased from 8.6 mV to 12.0 mV. A separate summary plot including Trial E showed a polynomial relationship between mean ROI echogenicity and MVIC with adjusted $R^2 = 0.9197$. This result supports the hypothesis that ultrasound signal changes within the muscle ROI can reflect fatigue-related decline in force capacity.

Trial	MVIC	Mean ROI Echogenicity
A	45	8.6 mV
B	43	8.7 mV
C	37	9.9 mV
D	32	12.0 mV
E	36	9.6

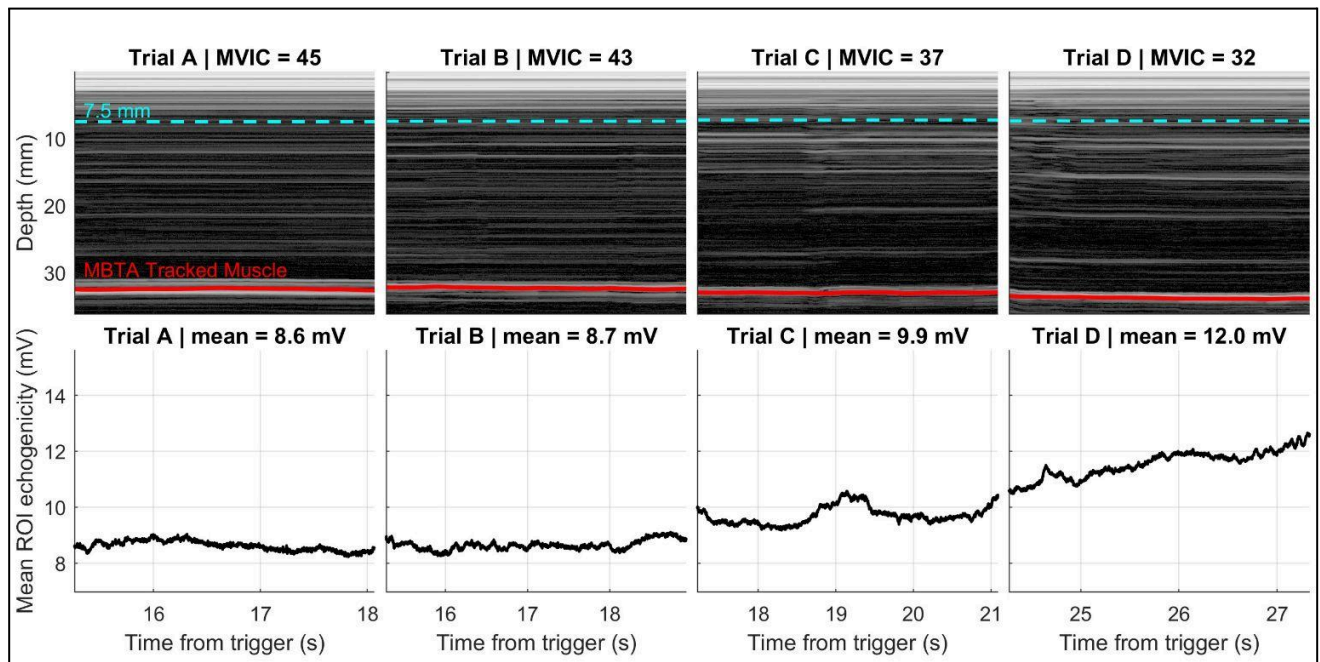


Figure 7. Echogenicity across trials from.

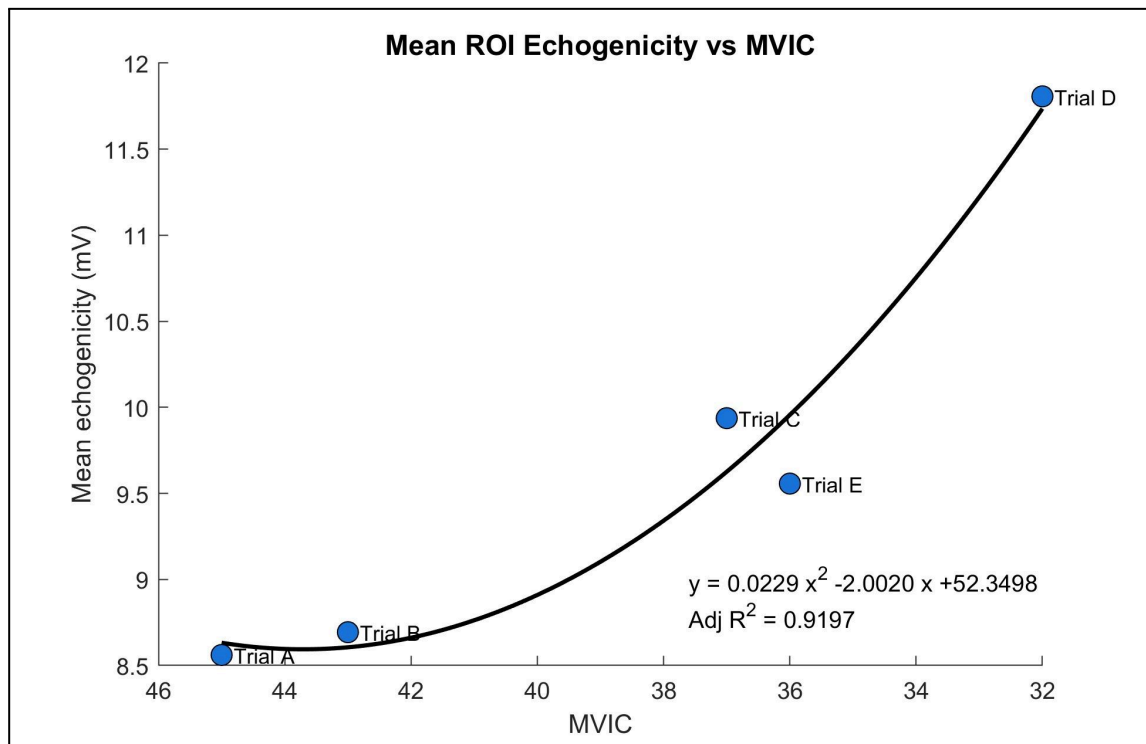


Figure 8. Mean ROI echogenicity vs. MVIC with fit and adjusted R^2 .

5.3 Electromechanical Delay Results

The EMD results examined whether ultrasound-derived mechanical delay followed the same directional fatigue trend as the EMG-to-torque baseline. In the within-trial visualizations, the detected onset bands became wider across repeated trials, suggesting increased delay as the muscle fatigued. Across subjects, the final presentation reports that Armando and Edgar showed the expected pattern: as fatigue increased and MVIC

percentage declined, both baseline EMG-to-torque delay and target EMG-to-ultrasound delay increased or followed consistent directional trends. Lorena showed an unexpected reversal between the two plots, which the team attributed to possible inaccuracies in A-mode ultrasound boundary tracking and onset detection.

Subject / Series	EMG-to-Torque R ²	EMG-to-US R ²	Interpretation
Lorena	0.71	0.55	Trend inconsistent with expected relationship; likely needs boundary/onset debugging.
Armando	0.57	0.45	Expected directional relationship in final presentation notes.
Edgar	0.85	0.84	Strongest evidence that ultrasound EMD tracks baseline delay.

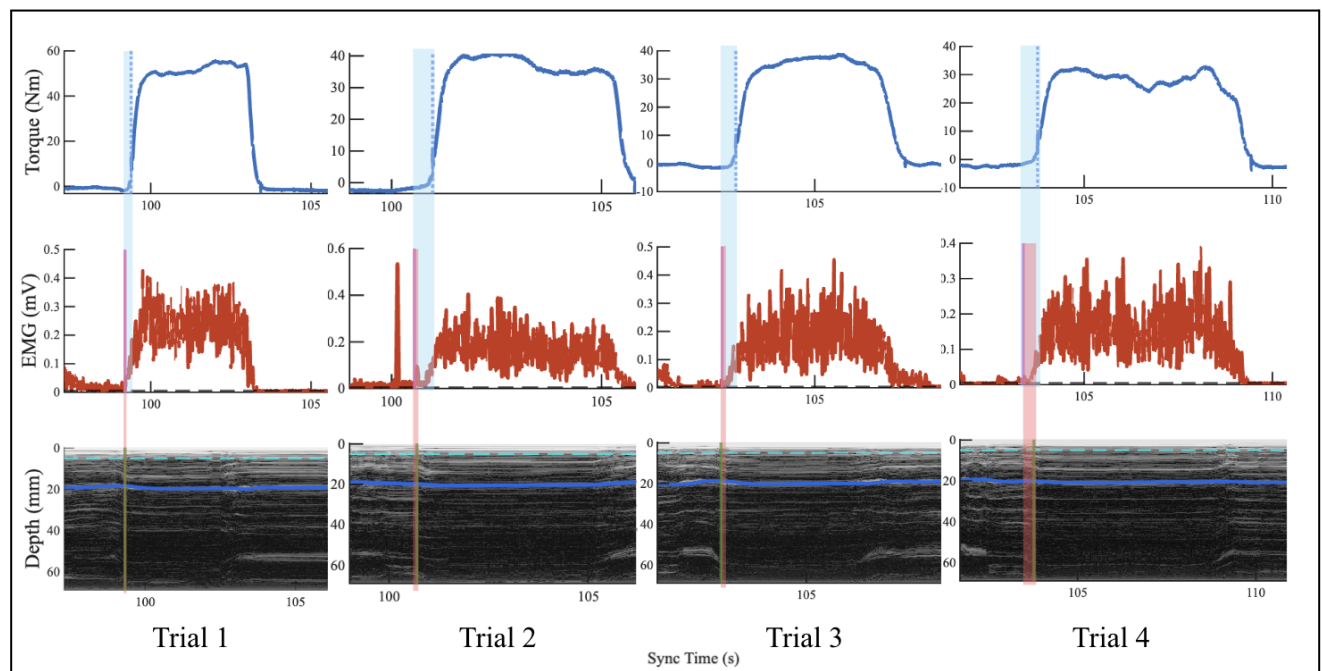


Figure 9. EMD across trials visualization.

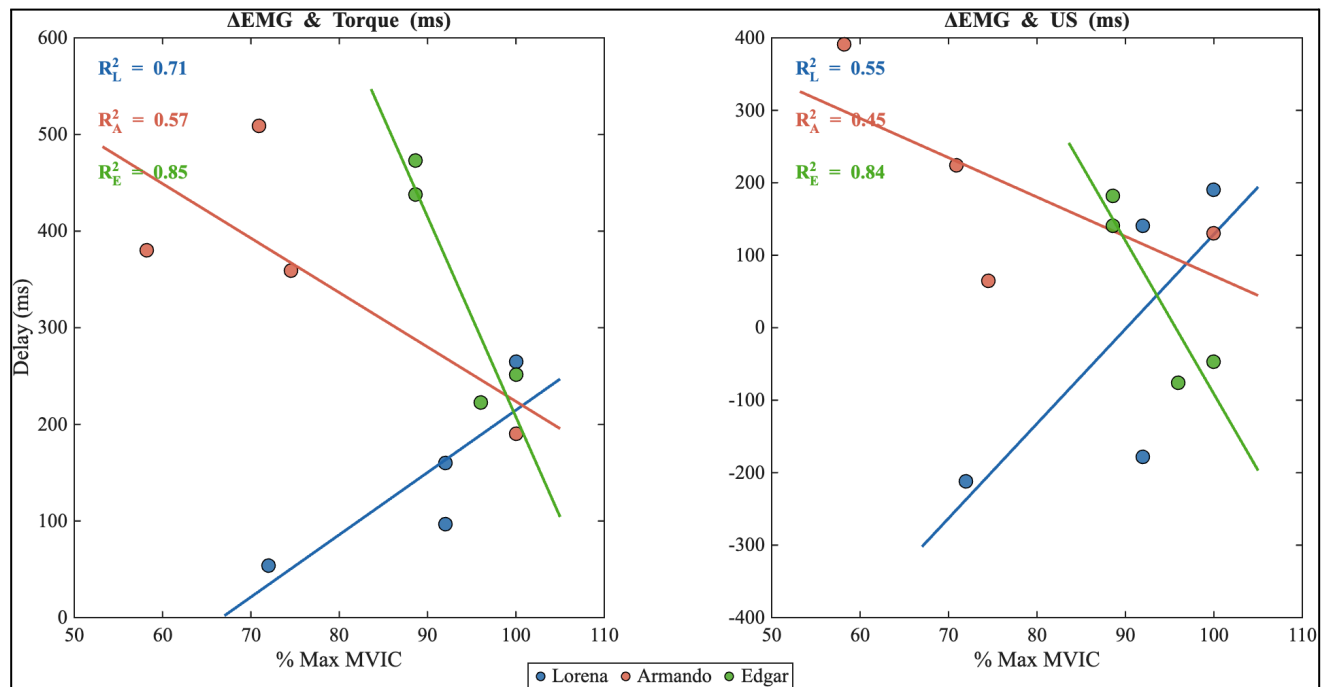


Figure 10. EMD across subjects comparing EMG-to-torque and EMG-to-ultrasound trends.

5.4 Validation Summary

Taken together, the verification work supports the feasibility of the proposed sensing approach but does not yet fully validate a wearable fatigue monitor. The strongest evidence is that ultrasound features changed with MVIC across trials and that the EMD pipeline produced physiologically plausible trends in two of three subjects. The main remaining uncertainty is robustness: performance depends on synchronization, boundary tracking, ROI selection, sensor placement, and subject-to-subject variability. Therefore, the most defensible final claim is that the team developed and tested algorithmic pipelines that demonstrate promising ultrasound- and EMG-derived fatigue features, not that the team has completed a fully validated clinical device.

6. Risk, Safety, Ethics, and Implementation Considerations

6.1 Risk Analysis

Failure Mode / Risk	Potential Effect	Mitigation
Probe shifts during contraction	ROI no longer represents the target muscle, producing false echogenicity or EMD trends.	Use mechanical fixation, placement guides, MBTA tracking, and signal-quality checks.
Excessive probe pressure	Compresses tissue and changes ultrasound signal independent of fatigue.	Limit strap pressure, design compliant mounting, verify pressure sensitivity.
EMG noise or electrode placement error	False onset detection or inaccurate EMD.	Use placement training, filtering, backward-from-peak onset detection, and quality metrics.
Synchronization error	Incorrect delay estimates across EMG, ultrasound, and torque.	Use hardware trigger, inspect alignment, quantify latency, and correct clock offsets.

Overinterpretation of small dataset	Unsupported claims about generalizability or injury prevention.	State limitations, collect more subjects, validate across tasks and populations.
User misreads fatigue zones	Incorrect training or rehab decisions.	Use clear labeling, confidence indicators, and clinician/trainer guidance.

6.3 Implementation Pathway

The long-term implementation pathway has three stages. First, refine the algorithms using more synchronized ultrasound, EMG, and torque data. Second, reduce the system to the minimum sensing and compute package needed for reliable fatigue estimates. Third, integrate the algorithms into a wearable form factor using compact ultrasound and EMG hardware. We identified three pathways: sports medicine, rehabilitation, and work ergonomics. Each pathway would require its own protocol and validation standard because movement patterns, fatigue thresholds, and acceptable risk levels differ across contexts.

7. Discussion

7.1 Key Contributions

This project contributed a structured algorithmic approach for measuring peripheral muscle fatigue using sparse ultrasound and EMG. The team moved from a broad wearable sensor concept to a more defensible signal-processing framework: local ultrasound was used to measure fatigue-related mechanical tissue response, EMG was used to measure activation timing, and torque/MVIC was used as the validation benchmark. This approach was physiology-driven because echogenicity and EMD were selected for their relationship to tissue backscatter, muscle deformation, and activation-to-mechanical delay. The project also produced working preliminary pipelines for both echogenicity and EMD: the echogenicity pipeline compared MBTA-bounded mean ROI intensity across declining MVIC trials, while the EMD pipeline synchronized torque, EMG, and ultrasound streams to support modality-specific onset detection. By validating these features against mechanical output rather than relying on sensor data alone, the project established a foundation for a future wearable fatigue-monitoring system that uses lower-data sparse ultrasound and EMG to generate an actionable fatigue index.

7.2 Limitations

The most important limitation is sample size. We report EMD across three subjects, and some echogenicity analyses appear to use a small number of repeated trials from one subject. This is sufficient for algorithm feasibility but not for generalizable validation. A second limitation is that the protocol remains laboratory-based and uses a dynamometer, PowerLab, and separate EMG and ultrasound systems. A third limitation is sensor robustness: the approach is sensitive to probe placement, mechanical perturbation, ROI choice, and boundary tracking. Finally, fatigue-zone classification is presented as the long-term output, but the uploaded materials do not yet show final classifier accuracy, macro-F1, or cross-validation results.

7.3 Future Work

Future work should first focus on strengthening the reliability of the measurement pipeline before expanding the dataset. The highest priority is to standardize the experimental protocol, including rest periods, trial timing, contraction instructions, and sensor placement, because uncontrolled recovery or setup

variability could obscure fatigue trends and require data to be recollected. In parallel, the MBTA and ROI-selection pipeline should be refined to reduce boundary-tracking errors and improve the reliability of ultrasound-derived echogenicity features. The team should also quantify synchronization latency between EMG, ultrasound, and torque signals, since uncorrected timing offsets would directly affect EMD calculations. Once these methodological issues are locked in, the next step is to collect a larger dataset across more subjects, days, and fatigue protocols to determine whether the observed trends generalize. With a cleaner and larger dataset, future work can then train and validate a fatigue-zone classifier that converts echogenicity and EMD features into an actionable fatigue output. Finally, the system should be translated into a wearable form factor by designing a fixation mechanism and placement guide that allow repeatable sensor positioning outside the laboratory.

8. Conclusion

This project developed a physiology-driven fatigue assessment pipeline that uses sparse ultrasound and EMG to quantify changes in local muscle behavior during repeated biceps contractions. The work addresses the need for objective, actionable fatigue monitoring in athletic training and rehabilitation settings. The final prototype used synchronized ultrasound, EMG, and torque data to evaluate two complementary fatigue features: echogenicity changes within a tracked muscle ROI and electromechanical delay between activation and mechanical response. Echogenicity results showed a strong relationship between signal and declining MVIC, while EMD results showed promising agreement with the EMG-to-torque baseline in two of three subjects. These findings suggest that sparse ultrasound and EMG can capture meaningful fatigue-related signals, but additional data, improved boundary tracking, controlled protocols, and wearable integration are needed before the system can support generalizable fatigue-zone feedback in real-world use.

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